Limits-1

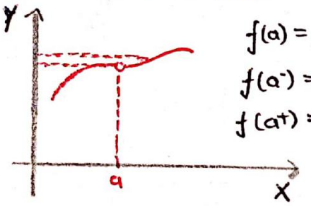
→ There are 7 indeterminate values at which exact value of function can't be calculated but its limiting value can be found.

$\left[\frac{0}{0}, \frac{\infty}{\infty}, 0\times\infty, \infty-\infty, 0^0, \infty^0, 1^\infty\right]$

↳ (7 Indeterminate Forms)

→ At a point $x=a$, the limit of function $f(x)$ can exist irrespective of whether $f(x)$ is defined at $x=a$ or not.

i.e



$f(a)$ = Not defined

$f(a^-)$ = exist

$f(a^+)$ = exist

Methods to solve $(0/0)$ and (∞/∞) forms

Factorization | Rationalisation | L' Hopital Rule (Differentiation)

Standard Forms :

1) $\lim_{\square\to 0} \frac{\ln(1+\square)}{\square} = 1$ ← Apply this formula directly only when entire '□' is tending to zero

i.e $\lim_{x\to 0} \frac{\ln(1+\sin x)}{\sin x} = 1$

But, $\lim_{x\to 0} \frac{\ln(1+\cos x)}{\cos x} \neq 1$ (Because, when $x\to 0 \Rightarrow \cos x \to 1$)

$= \ln 2$ (✓)

2) $\lim_{\square\to 0} \frac{a^\square - 1}{\square} = \ln a$

3) $\lim_{\square\to 0} \frac{e^\square - 1}{\square} = 1$

4) $\lim_{\square\to 0} \frac{1-\cos(\square)}{\square^2} = \frac{1}{2}$

Similarly, use these formula only when entire '□' tend to zero.

5) $\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \frac{x}{\sin x} = 1$

6) $\lim_{x\to 0} \frac{\tan x}{x} = \lim_{x\to 0} \frac{x}{\tan x} = 1$

7) $\lim_{x\to 0} \frac{\sin^{-1}x}{x} = \lim_{x\to 0} \frac{x}{\sin^{-1}x} = 1$

8) $\lim_{x\to 0} \frac{\tan^{-1}x}{x} = \lim_{x\to 0} \frac{x}{\tan^{-1}x} = 1$

Ques $\lim_{x\to 0} \left[\left(\frac{1-\cos^2(3x)}{\cos^3(4x)}\right)\left(\frac{\sin^3(4x)}{(\ln(2x+1))^5}\right)\right]$ is ____.

Sol^n $\lim_{x\to 0} \frac{\sin^2(3x)}{(3x)^2}\times\frac{\sin^3(4x)}{(4x)^3}\times\frac{1}{\cos^3(4x)}\times\frac{1}{(\ln(2x+1))^5}\times\frac{1}{(2x)^5}$

(each ratio $\to 1$)

$= \lim_{x\to 0} \frac{(3x)^2\cdot(4x)^3}{(2x)^5} \Rightarrow \frac{3^2\times 2^6}{2^5} \Rightarrow 18$ Ans

Ques The value of $\lim_{x\to 1} \frac{(x^2-1)\sin^2(\pi x)}{x^4-2x^3+2x-1}$ is ____.

Sol^n $= \lim_{x\to 1} \frac{(x^2-1)\left[\frac{\sin^2(\pi-\pi x)}{(\pi-\pi x)^2}\right](\pi-\pi x)^2}{x^4-2x^3+2x-1}$

Concept: Whenever there is $\sin\theta$, write it as $\sin(\pi-\theta)$

$= \lim_{x\to 1} \frac{(x^2-1)(\pi-\pi x)^2}{x^4-2x^3+2x-1}$

$x^4-x^3-x^3+x+x-1$

$= x^3(x-1) - x(x-1)(x+1) + 1(x-1)$

$= (x-1)(x^3-x^2+1) \Rightarrow (x-1)(x^2(x-1) - 1(x-1))$

$= (x-1)(x-1)(x^2-1)$

$= \lim_{x\to 1} \frac{(x^2-1)\pi^2(1-x)^2}{(x-1)(x-1)(x^2-1)} = \pi^2$ Ans

Ques $\lim_{x\to \frac{\pi}{4}} \frac{8\sqrt{2} - (\cos x+\sin x)^7}{\sqrt{2}-\sqrt{2}\sin 2x}$ is ____.

Sol^n It is $(0/0)$ form, so we apply LH-Rule

$= \lim_{x\to\pi/4} \frac{-7(\cos x+\sin x)^6(-\sin x+\cos x)}{-2\sqrt{2}\cos 2x}$

↳ $\cos^2x - \sin^2x = (\cos x-\sin x)\times(\cos x+\sin x)$

$= \lim \frac{-7\left(\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\right)^6}{-2\sqrt{2}\left(\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\right)}$

$= \frac{7}{2\sqrt{2}}(\sqrt{2})^4(\sqrt{2}) \Rightarrow 14$ Ans

Ques $\lim_{x\to\frac{\pi}{2}} \tan^2x\left(\sqrt{2\sin^2x+3\sin x-4} - \sqrt{\sin^2x+6\sin x+2}\right)$

Sol^n Since entire function is in 'sin', thus we convert $\tan^2x$ also into 'sin'

$= \lim_{x\to\frac{\pi}{2}} \frac{\sin^2x}{1-\sin^2x}\left(\sqrt{2\sin^2x+3\sin x-4} - \sqrt{\sin^2x+6\sin x+2}\right)$

Let, $\sin x = t$, thus when $(x\to\pi/2 \Rightarrow t\to 1)$

So, $= \lim_{t\to 1} \frac{t^2\left(\sqrt{2t^2+3t-4} - \sqrt{t^2+6t+2}\right)}{1-t^2}$

After Rationalising

$= \lim_{t\to 1} \frac{t^2(t-1)(t-2)}{-(t-1)(t+1)\left(\sqrt{2t^2+3t-4}+\sqrt{t^2+6t+2}\right)}$

$= \frac{1(1-2)}{-(1+1)(3+3)} \Rightarrow \frac{1}{12}$ Ans

Method to Solve $(\infty-\infty)$ & $(0\times\infty)$:

Step-01 ⇒ Convert the given forms into $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form by doing substitution.

$\lim_{x\to\infty}$ □ $(x=\frac{1}{t})$ → $\lim_{t\to 0}$ □

$\lim_{x\to-\infty}$ □ $(x=-\frac{1}{t})$ → $\lim_{t\to 0}$ □

Step-02 ⇒ Use factorization / Rationalisation / LH-Rule to solve further.

Ques $\lim_{x\to\infty} \left(\sqrt{25x^2-3x} - 5x\right)$

Sol^n It is $(\infty-\infty)$ form, so we put $x=\frac{1}{t}$

$= \lim_{t\to 0}\left(\sqrt{\frac{25}{t^2}-\frac{3}{t}} - \frac{5}{t}\right)$

$= \lim_{t\to 0}\left(\frac{\sqrt{25-3t}-5}{t}\right)\times\frac{\sqrt{25-3t}+5}{\sqrt{25-3t}+5}$

$= \lim_{t\to 0} \frac{-3t}{t(\sqrt{25-3t}+5)} \Rightarrow \frac{-3}{10}$ Ans

Ques If $\lim_{n\to\infty}\left(\sqrt{n^2-n-1} + n\alpha+\beta\right) = 0$ then $8(\alpha+\beta)$ is equal to ____.

Sol^n Put, $n=1/t$

$= \lim_{t\to 0}\left(\sqrt{\frac{1}{t^2}-\frac{1}{t}-1} + \frac{\alpha}{t}+\beta\right)$

$= \lim_{t\to 0}\left(\frac{\sqrt{1-t-t^2}+\alpha}{t}\right)+\beta$

Now, we are given the value of limit is '0' i.e limit do exist.

For limit to exist, the expression $\left(\frac{\sqrt{1-t-t^2}+\alpha}{t}\right)$ should attain $\frac{0}{0}$ form after substituting $n=1/t$. So, value of α should be such that after putting $t\to 0$, the entire expression becomes $(0/0)$

i.e $\frac{\sqrt{1-0-0}+\alpha}{0} = \frac{0}{0} \Rightarrow \frac{1+\alpha}{0} = \frac{0}{0}$

$\Rightarrow \boxed{\alpha=-1}$

Now,

$= \lim_{t\to 0}\left(\frac{\sqrt{1-t-t^2}-1}{t}+\beta\right) = 0$

$= \lim_{t\to 0}\left(\frac{1-t-t^2-1}{t(\sqrt{1-t-t^2}+1)}+\beta\right) = 0$

$= \frac{-1}{2}+\beta = 0 \Rightarrow \boxed{\beta=\frac{1}{2}}$

Hence, $8(\alpha+\beta) = 8\left(-1+\frac{1}{2}\right) \Rightarrow -4$ Ans

Binomial Approximation :

It is applicable only when

(i) First term = 1

(ii) The value of □ → 0

$$\boxed{(1+\square)^n \simeq 1+n\square+\frac{n(n-1)}{2}\square^2}$$

Generally we use only till $(1+n\square)$

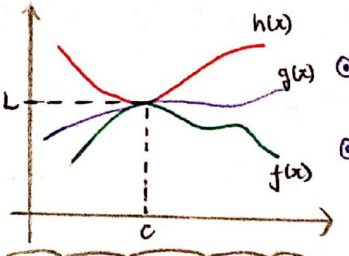
Ques The value of $\lim_{x\to 0}\left(\frac{x}{\sqrt[3]{1-\sin x}-\sqrt[3]{1+\sin x}}\right)^{3x}$ is ____.

Sol^n Since $x\to 0 \Rightarrow (\sin x)\to 0$, so we can use Binomial approximation here.

$= \lim_{x\to 0} \frac{x}{\left(x-\frac{1}{6}\sin x\right)-\left(x+\frac{1}{6}\sin x\right)}$

$= \lim_{x\to 0} \frac{6x}{-2(\sin x)} \Rightarrow -4$ Ans

Sandwich / Squeeze Theorem :



If,

① $f(x) \le g(x) \le h(x)$

And,

② $\lim_{x\to c} f(x) = \lim_{x\to c} h(x) = L$

Then,

$\lim_{x\to c} f(x) = \lim_{x\to c} g(x) = L$

Agar upar aur neeche wale function KP limit same hai toh beech wale function KP bhi same limit hogi.

We use this theorem mainly for 4 types of functions.

Function having {x}

Use $0 \le \{x\} < 1$

Function having [x]

Use $x-1 < [x] \le x$

Algebraic / Trigonometric / Variable in fraction

For lesser function, take min. value of T-ratio; for higher function, take max. value of T-ratio

Eg: $\lim_{x\to\infty} \frac{\{x\}}{x}$

Sol: $0 \le \{x\} < 1 \Rightarrow \frac{0}{x} \le \frac{\{x\}}{x} < \frac{1}{x}$

$\lim_{x\to\infty} \frac{0}{x} \le \frac{\{x\}}{x} < \lim_{x\to\infty} \frac{1}{x}$

$0 \qquad\qquad 0$

$\Rightarrow \lim_{x\to\infty} \frac{\{x\}}{x} = 0$ Ans

Eg: $\lim_{x\to\infty} \frac{[x]}{x}$

Sol: $x-1 < [x] \le x \Rightarrow \frac{x-1}{x} < \frac{[x]}{x} \le \frac{x}{x}$

$1-\frac{1}{x} < \frac{[x]}{x} \le 1 \Rightarrow \lim_{x\to\infty} \frac{[x]}{x} = 1$ Ans

$\lim_{x\to\infty}\left(1-\frac{1}{x}\right) = 0 = 1$

Ques $\lim_{x\to\infty} \frac{x+7\sin x}{-2x+13}$

Sol: $\frac{x-7}{-2x+13} \le \frac{x+7\sin x}{-2x+13} \le \frac{x+7}{-2x+13}$

(We had put $\sin x = -1$) (We had put $\sin x = +1$)

$\lim_{x\to\infty} \frac{x-7}{-2x+13} = \lim_{x\to\infty} \frac{x+7}{-2x+13} = -\frac{1}{2}$

So, $\lim_{x\to\infty} \frac{x+7\sin x}{-2x+13} = -\frac{1}{2}$ Ans

Polynomial Limits (Only when $x\to\infty$)

Let $y = \lim_{x\to\infty} \frac{g(x)}{f(x)}$ where $g(x)$ & $f(x)$, both are polynomial function.

Now, if

Degree of $g(x)$ > Degree of $f(x) \Rightarrow y = \infty$

Degree of $g(x)$ < Degree of $f(x) \Rightarrow y = 0$

Degree of $g(x)$ = Degree of $f(x) \Rightarrow y$ = Ratio of leading coeff of $g(x)$ & $f(x)$

Ques $\lim_{n\to\infty} \frac{1}{1+n^2} + \frac{2}{2+n^2} + \cdots \frac{n}{n+n^2}$

Sol: Let, $\frac{1}{1+n^2} + \frac{2}{2+n^2} + \cdots \frac{n}{n+n^2} = g(n)$

Now,

$\underbrace{\frac{1}{n+n^2} + \frac{2}{n+n^2} \cdots \frac{n}{n+n^2}}_{\text{Largest term in denominator}} \le g(n) \le \underbrace{\frac{1}{1+n^2} + \frac{2}{n+1} \cdots \frac{n}{1+n^2}}_{\text{Smallest term in denominator}}$

$\lim_{n\to\infty}\left(\frac{\frac{n(n+1)}{2}}{n+n^2}\right) = \frac{1}{2} \Rightarrow \frac{1}{2}$

$\lim_{n\to\infty}\left(\frac{\frac{n(n+1)}{2}}{1+n^2}\right) = \frac{1}{2} \Rightarrow \frac{1}{2}$

Thus, $\lim_{n\to\infty} g(n) = \frac{1}{2}$ Ans

Important Expansions :

1) $a^x = 1 + \frac{x\ln a}{1!} + \frac{x^2\ln^2 a}{2!} + \frac{x^3\ln^3 a}{3!} \cdots$

2) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

3) $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$

Trigonometric Expansions ⇒

1) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$

2) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$

3) $\tan x = x + \frac{x^3}{3!} + \frac{2x^5}{15} + \cdots$

Inverse Trigonometric Expansions ⇒

1) $\sin^{-1}x = x + \frac{1^2}{3!}x^3 + \frac{1^2\cdot 2^2}{5!}x^5 + \cdots$

2) $\cos^{-1}x = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \frac{61x^6}{6!} + \cdots$

3) $\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$

Ques If value of $\lim_{x\to 0} \frac{\sin^{-1}x - \tan^{-1}x}{3x^3}$ is L, then find $(6L+1)$.

Sol: $\lim_{x\to 0} \frac{\left(x + \frac{x^3}{3!}\right) - \left(x - \frac{x^3}{3}\right)}{3x^3}$

$\lim_{x\to 0} \frac{\frac{1}{6} + \frac{1}{3}}{3} \Rightarrow L = 1/6$

So, $6L+1 = 2$ Ans

Ques Let $\beta = \lim_{x\to 0} \frac{\alpha x - (e^{3x}-1)}{\alpha x(e^{3x}-1)}$ for some $\alpha \neq R$, then find value of $(\alpha+\beta)$.

Sol: $\beta = \lim_{x\to 0} \frac{\alpha x - (e^{3x}-1)}{\alpha x \underbrace{\left(\frac{e^{3x}-1}{3x}\right)}_{g(x)} 3x}$ $\Rightarrow \lim_{x\to 0} g(x) = 1$

$\beta = \lim_{x\to 0} \frac{\alpha x - \left(1 + \frac{3x}{1!} + \frac{(3x)^2}{2!}\right) + 1}{3\alpha x^2}$

$\beta = \lim_{x\to 0} \frac{\alpha x - 3x - \frac{9x^2}{2}}{3\alpha x^2}$

$\beta = \lim_{x\to 0} \frac{\alpha - 3}{3\alpha x} + \lim_{x\to 0} \frac{-9}{6\alpha}$

↳ For this limit to exist, function must take $(0/0)$ form, so

$\Rightarrow \alpha - 3 = 0 \Rightarrow \alpha = 3$

By applying LH rule

$\lim_{x\to 0} \frac{\alpha - 3}{3\alpha x} \Rightarrow 0$

So, $\beta = \lim_{x\to 0} \frac{-9x}{2\cdot 6(3)} \Rightarrow \beta = -\frac{1}{2}$

Hence, $\alpha+\beta = 3 - \frac{1}{2} \Rightarrow \frac{6-1}{2}$

$\alpha+\beta = \frac{5}{2}$ Ans

Ques If $\lim_{x\to 0} \frac{e^{ax} - \cos(bx) - \frac{cxe^{-ex}}{2}}{1-\cos(2x)} = 17$

then $5a^2+b^2$ is equal to ____.

Sol: Here, we need to use expansions and binomial approximation to solve the ques.

$\lim_{x\to 0} \frac{\left(1 + ax + \frac{a^2x^2}{2}\right) - \left(1 - \frac{b^2x^2}{2}\right) - \frac{cx}{2}(1-cx)}{(1) \Leftarrow \left(\frac{1-\cos(2x)}{(2x)^2}\right) \times (2x)^2}$

$\lim_{x\to 0} \frac{\left(a - \frac{c}{2}\right)x + \left(\frac{a^2}{2} + \frac{b^2}{2} + \frac{c^2}{2}\right)x^2 = 17}{2x^2}$

$\lim_{x\to 0} \frac{\left(a-\frac{c}{2}\right)}{2x} + \lim_{x\to 0} \frac{\left(\frac{a^2}{2} + \frac{b^2}{2} + \frac{c^2}{2}\right)}{2} = 17$

Limits - 2

Now for, $\lim_{x\to 0} \frac{(a-\frac{c}{2})}{2x}$ to exist, it must convert int $(0/0)$ form.

$\Rightarrow a-\frac{c}{2}=0 \Rightarrow c=2a$

$\boxed{c^2=4a^2}$

Also, by LH rule

$$\boxed{\lim_{x\to 0}\left(\frac{a-\frac{c}{2}}{2x}\right)=0}$$

So,

$$\lim_{x\to 0} a^2+b^2+c^2=68$$

$$a^2+b^2+4a^2=68 \Rightarrow \boxed{5a^2+b^2=68}$$ Ans

Method to Solve 0^0 / ∞^0 form :

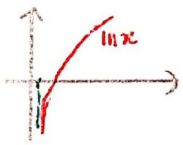
↳ Take log on both sides after assuming limit equal to 'y'.

Eg: $\lim_{x\to 0} x^x$

Sol^n Let, $y=\lim_{x\to 0} x^x \Rightarrow \ln y = \lim_{x\to 0} x\cdot\ln x$

$\Rightarrow \ln y = \lim_{x\to 0} \frac{\ln x}{(1/x)}$ ← (Got converted to $(\frac{-\infty}{\infty})$ form bcz when $x\to 0$, $\ln x\to -\infty$)

As, $x\to 0$, $\ln x\to -\infty$



So here, we can simply apply LH rule ⇒

$$\ln y = \lim_{x\to 0}\left(\frac{1/x}{-1/x^2}\right) \Rightarrow \lim_{x\to 0}\frac{-x^2}{x}$$

$$\ln y = \lim_{x\to 0} -x \Rightarrow 0$$

$\boxed{y=e^0\Rightarrow 1}$ Ans

Eg: $\lim_{x\to 0} x^{\tan x}$

Sol^n $y=\lim_{x\to 0} x^{\tan x} \Rightarrow \ln y = \lim_{x\to 0} \tan x\cdot\ln x$

$$\ln y = \lim_{x\to 0}\frac{\ln x}{\cot x}$$

Basically, we need to manipulate function such that we get $\frac{\pm\infty}{\infty}$ or $\frac{0}{0}$ form

Apply LH rule

$$\ln y = \lim_{x\to 0}\frac{1/x}{-\operatorname{cosec}^2 x} \Rightarrow \lim_{x\to 0}\frac{-\sin^2 x}{x\times x}\times x$$

$$\ln y = \lim_{x\to 0} -x \Rightarrow \ln y = 0 \Rightarrow \boxed{y=1}$$ Ans

Ques $\lim_{x\to 0}\left(\frac{1}{x}\right)^{1-\cos x}$

Sol^n $y=\lim_{x\to 0}\left(\frac{1}{x}\right)^{1-\cos x}$

$$\ln y = \lim_{x\to 0}(1-\cos x)\ln\left(\frac{1}{x}\right)$$

$$\ln y = \lim_{x\to 0}\left(\frac{1-\cos x}{x^2}\right)\cdot x^2\cdot(-\ln x)$$

$$\ln y = \frac{1}{2}\lim_{x\to 0}\frac{-\ln x}{1/x^2}$$

Applying LH rule

$$\ln y = \frac{1}{2}\lim_{x\to 0}\frac{-1/x}{-2/x^3} \Rightarrow \frac{1}{2}\lim_{x\to 0}\frac{x^2}{2}$$

$\ln y = 0 \Rightarrow \boxed{y=1}$ Ans

Method to solve 1^∞ form :

Whenever a function like $(f(x))^{g(x)}$ takes up 1^∞ form upon approaching its limiting value, we can directly write it as

V. Imp

$$\boxed{\lim_{x\to a}(f(x))^{g(x)} = e^{\lim_{x\to a}(f(x)-1)g(x)}}$$

Ques If $\alpha=\lim_{x\to\frac{\pi}{4}}\frac{\tan^3 x-\tan x}{\cos(x+\pi/4)}$ and $\beta=\lim_{x\to 0}(\cos x)^{\cot x}$ are roots of the equation $ax^2+bx-4=0$, then the ordered pair (a,b) is ____.

Sol^n $\alpha=\lim_{x\to\frac{\pi}{4}}\frac{\tan^3 x-\tan x}{\cos(x+\pi/4)}$ $\left(\frac{0}{0}\right)$

By LH rule

$$\alpha=\lim_{x\to\frac{\pi}{4}}\frac{3\tan^2 x\cdot\sec^2 x-\sec^2 x}{-\sin(x+\pi/4)}$$

$$\alpha=\frac{(3\times 1\times 2)-2}{-1} \Rightarrow \boxed{\alpha=-4}$$

Now, $\beta=\lim_{x\to 0}(\cos x)^{\cot x}$ (1^∞)

$$\Rightarrow \beta = e^{\lim_{x\to 0}(\cos x-1)\cot x}$$

$$\beta = e^{\lim_{x\to 0}-\left(\frac{1-\cos x}{x^2}\right)\cdot x^2\left(\frac{1}{\tan x}\cdot x\right)\frac{1}{x}}$$

($\frac{1-\cos x}{x^2}\to\frac{1}{2}$, $\frac{1}{\tan x}\cdot x\to 1$)

$$\beta = e^{-\frac{1}{2}\times 0\times 1\times 0} \Rightarrow e^0$$

$\boxed{\beta=1}$

Now, $ax^2+bx-4=0$ with $\alpha=-4$, $\beta=1$

Sum of roots $=\frac{-b}{a}=-3 \Rightarrow \boxed{b=3}$

Product of roots $=\frac{c}{a} \Rightarrow \frac{-4}{a}=-4 \Rightarrow \boxed{a=+1}$

Hence, $\boxed{(a,b)\equiv(1,3)}$ Ans

Limits involving Greatest Integer Functions :

→ When G.I.F is present, then always calculate Left Hand Limit (LHL) and Right Hand Limit (RHL) separately.

→ Simply substitute the value of (a^+) and (a^-) in $[x]$ and treat it as a constant value.

Ques Let 'a' be an integer such that $\lim_{x\to 7}\frac{18-[1-x]}{[x-3a]}$ exists. Then a is equal to ____.

Sol^n We can take the constant out of greatest integer function

$$=\lim_{x\to 7}\frac{18-([1]+[-x])}{[x]-3a}$$

LHL: $\lim_{x\to 7^-}\frac{18-1-[-6.999]}{[6.999]-3a}$

$=\frac{17-(-7)}{6-3a}$

$=\frac{24}{6-3a}$

RHL: $\lim_{x\to 7^+}\frac{18-1-[-7.011]}{[7.011]-3a}$

$=\frac{17-(-8)}{7-3a}$

$=\frac{25}{7-3a}$

For limit to exist,

$$LHL = RHL$$

$$\frac{24}{6-3a}=\frac{25}{7-3a}$$

$$56-24a=50-25a$$

$\boxed{a=-6}$ Ans

Ques The value of $\lim_{x\to 0^+}\frac{\cos^{-1}(x-[x]^2)\cdot\sin^{-1}(x-[x]^2)}{x-x^3}$ is ____.

Sol^n $\lim_{x\to 0^+}\frac{\cos^{-1}(x\to[0.111]^2)\cdot\sin^{-1}(x\to[0.111]^2)}{x-x^3}$ (with $x\to 0$)

$$=\lim_{x\to 0^+}\frac{\cos^{-1}x\cdot\sin^{-1}x}{x(1-x^2)}$$

$$=\underbrace{\lim_{x\to 0^+}\frac{\sin^{-1}x}{x}}_{1}\times\lim_{x\to 0^+}\frac{\cos^{-1}x}{1-x^2}$$

$$=\frac{\cos^{-1}(0)}{1-(0)^2} \Rightarrow \frac{\pi}{2}$$ Ans